

STA 2002

Lecture 17

Chi-square statistic

Testing goodness-of-fit with given class probabilities

Testing the goodness-of-fit with given parameter

Testing the goodness-of-fit with unknown parameter

Reminder – multinomial distribution

$$(X_1, \ldots, X_k) \sim \text{Multinomial}\,(n, \theta_1, \ldots, \theta_k)$$

a response s that takes k possible values $1, 2, \ldots, k$ — with the probability distribution given by $P(s = i) = \theta_i$

For a sample $(s_1, \ldots, s_n)$, define the counts $X_i = \sum_{j=1}^{n} I_{\{i\}}(s_j)$ for $i = 1, \ldots k$.

$$p_{(X_1,\ldots,X_k)}(x_1, \ldots, x_k) = \binom{n}{x_1 \ \ldots \ x_k} \theta_1^{x_1} \cdots \theta_k^{x_k}$$

$$x_i \in \{0, \ldots, n\} \text{ and } x_1 + \cdots + x_k = n.$$

$$\binom{n}{k_1\, k_2 \ \ldots \ k_l} = \frac{n!}{k_1!k_2! \cdots k_l!}$$

Reminder – multinomial distribution

$$
\begin{aligned}
p_{(X_1,X_2,X_3)}(x_1, x_2, x_3) &= \binom{n}{x_1}\binom{n-x_1}{x_2}\binom{n-x_1-x_2}{x_3}\theta_1^{x_1}\theta_2^{x_2}\theta_3^{x_3} \\
&= \binom{n}{x_1\, x_2\, x_3}\theta_1^{x_1}\theta_2^{x_2}\theta_3^{x_3}.
\end{aligned}
$$

As a simple example, suppose that we have an urn containing 10 red balls, 20 white balls, and 30 black balls. If we randomly draw 10 balls from the urn with replacement, what is the probability that we will obtain 3 red, 4 white, and 3 black balls? Because we are drawing with replacement, the draws are i.i.d., so the counts are distributed Multinomial(10, 10/60, 20/60, 30/60) . The required probability equals

$$
\binom{10}{3\,4\,3}\left(\frac{10}{60}\right)^3\left(\frac{20}{60}\right)^4\left(\frac{30}{60}\right)^3 = 3.0007 \times 10^{-2}.
$$

The chi-square statistic

Theorem 9.1.1 If $(X_1, \ldots, X_k) \sim \text{Multinomial}(n, p_1, \ldots, p_k)$, then

$$\sum_{i=1}^{k} \frac{(X_i - np_i)^2}{np_i} \xrightarrow{D} \chi^2 (k-1)$$

as $n \to \infty$.

We refer to X^2 as the *chi-squared statistic*.

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

O = observed frequency
E = expected frequency

k-1: only k-1 of the proportions can vary.

(Proof is too involved for this course, requires the derivation of $Cov(X_i, X_j)$ and matrix theories.

The chi-square statistic

A special case – the binomial distribution

Toss a coin with $P(head) = p_1, P(tail) = p_2$

$p_1 + p_2 = 1$

Find N_1 heads and N_2 tails, $N_1+N_2=n$

$$\chi^2 = \frac{(N_1 - np_1)^2}{np_1} + \frac{(N_2 - np_2)^2}{np_2}$$

$$= \frac{(N_1 - np_1)^2}{np_1} + \frac{(n - N_1 - n(1-p_1))^2}{n(1-p_1)}$$

$$= \frac{(N_1 - np_1)^2}{np_1} + \frac{(N_1 - np_1)^2}{n(1-p_1)}$$

$$= \left[\frac{N_1 - np_1}{\sqrt{np_1(1-p_1)}}\right]^2 \approx Z^2 \sim \chi_1^2.$$

The chi-square statistic

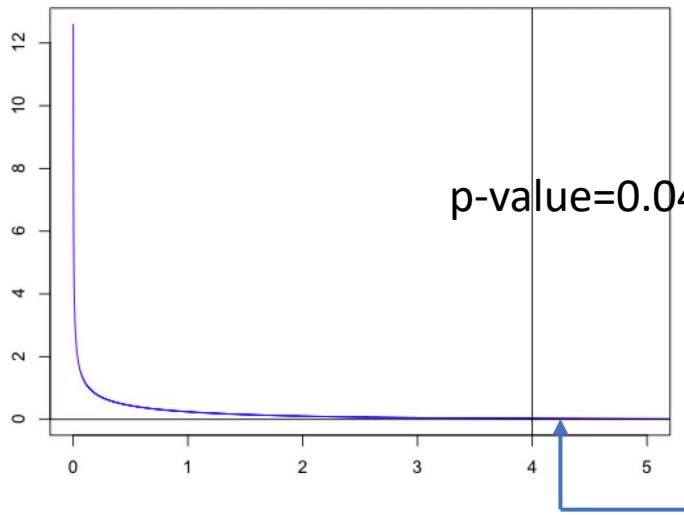
A coin toss with 40 heads and 60 tails
Assume this is a fair coin, i.e., $p = 0.5$

O	E	$O-E$	$(O-E)^2/E$
40	50	-10	2
60	50	10	2

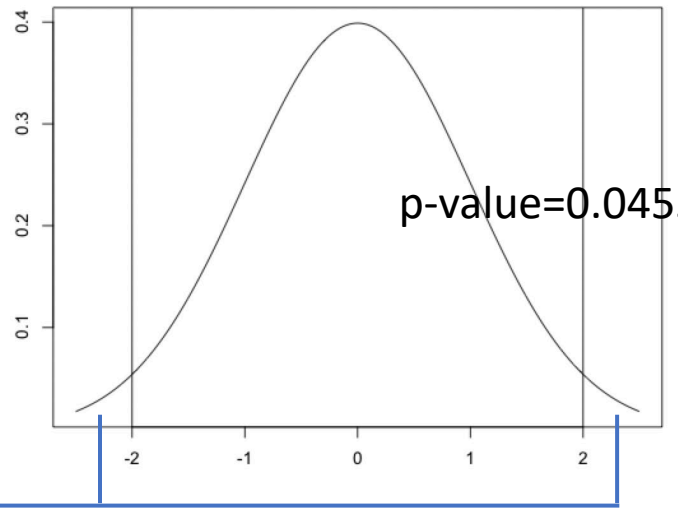
$$\chi^2 = \Sigma \frac{(O-E)^2}{E} = 4$$

$$df = 1$$

$$z = \frac{0.4 - 0.5}{\sqrt{0.5 \times 0.5/100}} = -2$$



p-value=0.0455



p-value=0.0455

Goodness-of-fit

$$f : [0, 1]^m \to [0, 1]$$

Testing the Accuracy of a Random Number Generator

$$U_{m+1} = f(U_1, \ldots, U_m)$$

In effect, every Monte Carlo simulation can be considered to be a set of mathematical operations applied to a stream of numbers $U_1, U_2, \ldots$ in [0, 1] that are supposed to be i.i.d. Uniform[0, 1].

$$U_{m+2} = f(U_2, \ldots, U_{m+1})$$

.....

How do we assess whether a particular *f* is good or not?

When the numbers follow uniform[0,1] is true, at least the numbers should fall evenly into 10 even-sized bins between 0 and 1.

This is only one criterion. Failing to reject doesn't mean the numbers are random i.i.d. from U[0,1].

i	x_i
1	993
2	1044
3	1061
4	1021
5	1017
6	973
7	975
8	965
9	996
10	955

$$\chi^2 = 11.056$$

$$\text{P-value } P(\chi^2 \geq 11.0560) = 0.27190 \text{ when } \chi^2 \sim \chi^2(9)$$

Null is not rejected.

Goodness-of-fit

If persons are asked to record a string of random digits, such as

3 7 2 4 1 9 7 2 1 5 0 8 . . . ,

The probability of the next digit being the same as the preceding one is $p_{10} = 1/10$,
The probability of the next being only one away from the preceding (assuming that 0 is one away from 9) is $p_{20} = 2/10$,
The probability of all other possibilities is $p_{30} = 7/10$.

$$H_0: \; p_1 = p_{10} = \frac{1}{10}, \; p_2 = p_{20} = \frac{2}{10}, \; p_3 = p_{30} = \frac{7}{10}$$

	Frequency	Expected Number
Same	0	50(1/10) = 5
One away	8	50(2/10) = 10
Other	42	50(7/10) = 35
Total	50	50

$$\frac{(0-5)^2}{5} + \frac{(8-10)^2}{10} + \frac{(42-35)^2}{35} = 6.8 > 5.991 = \chi^2_{0.05}(2)$$

Null is rejected. The digits don't appear random as assumed.

Goodness of fit when cell probabilities depend on a parameter

Toss 4 coins. Record the number of heads.

In 100 trials,

H_0: $X \sim Binomial(4, 1/2)$

# heads	Frequency	Expected under H_0
0	7	$100 \times \binom{4}{0}\left(\frac{1}{2}\right)^4\left(\frac{1}{2}\right)^0 = 6.25$
1	18	$100 \times \binom{4}{1}\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^1 = 25$
2	40	$100 \times \binom{4}{2}\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^2 = 37.5$
3	31	$100 \times \binom{4}{3}\left(\frac{1}{2}\right)^1\left(\frac{1}{2}\right)^3 = 25$
4	4	$100 \times \binom{4}{4}\left(\frac{1}{2}\right)^0\left(\frac{1}{2}\right)^4 = 6.25$

$$\frac{(7-6.25)^2}{6.25} + \frac{(18-25)^2}{25} + \frac{(40-37.5)^2}{37.5} + \frac{(31-25)^2}{25} + \frac{(4-6.25)^2}{6.25} = 4.47$$

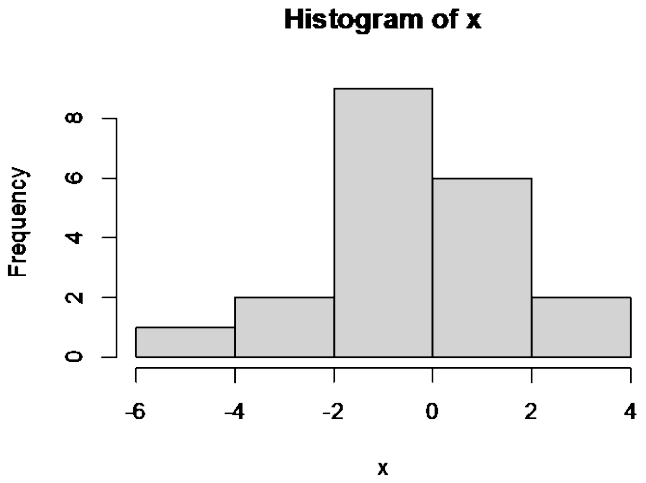
$$\chi^2_{0.05}(4) = 9.488$$

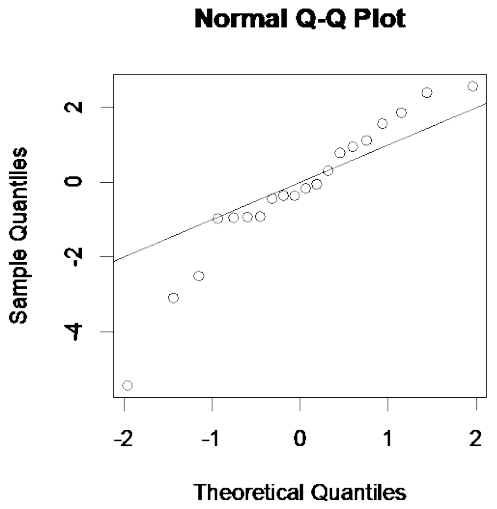
Null is not rejected.

Goodness of fit when cell probabilities depend on a parameter

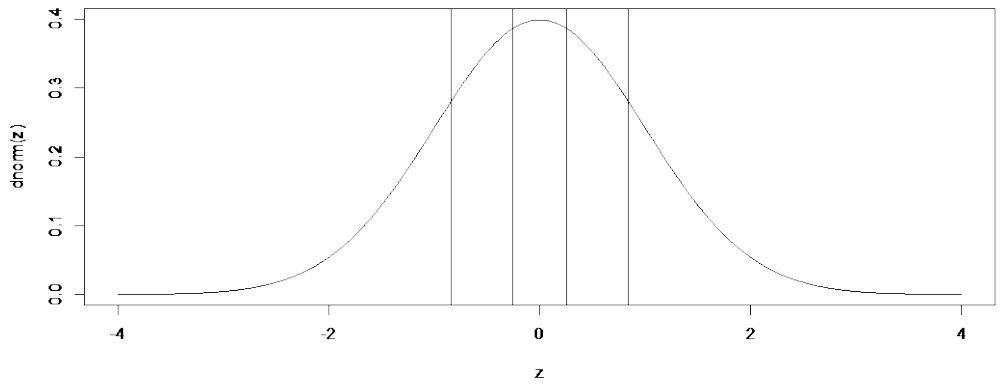
Did these number come from N(0,1)?

```
 2.40  0.79 -5.44 -0.16 -0.36 -0.97 -0.92 -0.05
-3.10  1.12 -2.51  1.57  0.95 -0.95  0.31 -0.44
 1.86  2.57 -0.93 -0.36
```





Five even-sized bins based on CDF of N(0,1)



0	E
7	4
3	4
2	4
2	4
6	4

$\chi^2 = 5.5$

$\chi^2_{4,\,0.05} = 9.49$

Null is not rejected.

Goodness of fit when cell probabilities depend on a parameter

Example 6.1-5 The following table lists 105 observations of X, the times in minutes between calls to 911:

30	17	65	8	38	35	4	19	7	14	12	4	5	4	2
7	5	12	50	33	10	15	2	10	1	5	30	41	21	31
1	18	12	5	24	7	6	31	1	3	2	22	1	30	2

Table 9.1-2 Summary of times between calls to 911

Class	Frequency	Probability	Expected
$A_1 = [0, 9]$	41	0.3624	38.0520
$A_2 = (9, 18]$	22	0.2311	24.2655
$A_3 = (18, 27]$	11	0.1473	15.4665
$A_4 = (27, 36]$	10	0.0939	9.8595
$A_5 = (36, 45]$	9	0.0599	6.2895
$A_6 = (45, 54]$	5	0.0382	4.0110
$A_7 = (54, 63]$	2	0.0244	2.5620
$A_8 = (63, 72]$	3	0.0155	1.6275
$A_9 = (72, \infty)$	2	0.0273	2.8665

H_0: the distribution of X is exponential with a mean of $\theta = 20$.

$$\frac{(41 - 38.0520)^2}{38.0520} + \frac{(22 - 24.2655)^2}{24.2655} + \cdots + \frac{(2 - 2.8665)^2}{2.8665} = 4.6861.$$

Checking χ^2_8 distribution,

p-value = 0.7905

PDF: $\frac{1}{\theta} e^{-x/\theta}$

CDF: $1 - e^{-x/\theta}$

Goodness of fit when cell probabilities depend on an **unknown** parameter

More generally, we will not have a prescribed probability distribution P for X but rather a statistical model $\{P_\theta : \theta \in \Omega\}$

Again we have $(X_1, \ldots, X_k) \sim \text{Multinomial}(n, p_1(\theta), \ldots, p_k(\theta))$

We can find MLE $\hat{\theta}$ from the likelihood function

$$L(\theta \mid x_1, \ldots, x_k) = (p_1(\theta))^{x_1} \cdots (p_k(\theta))^{x_k}$$

Then

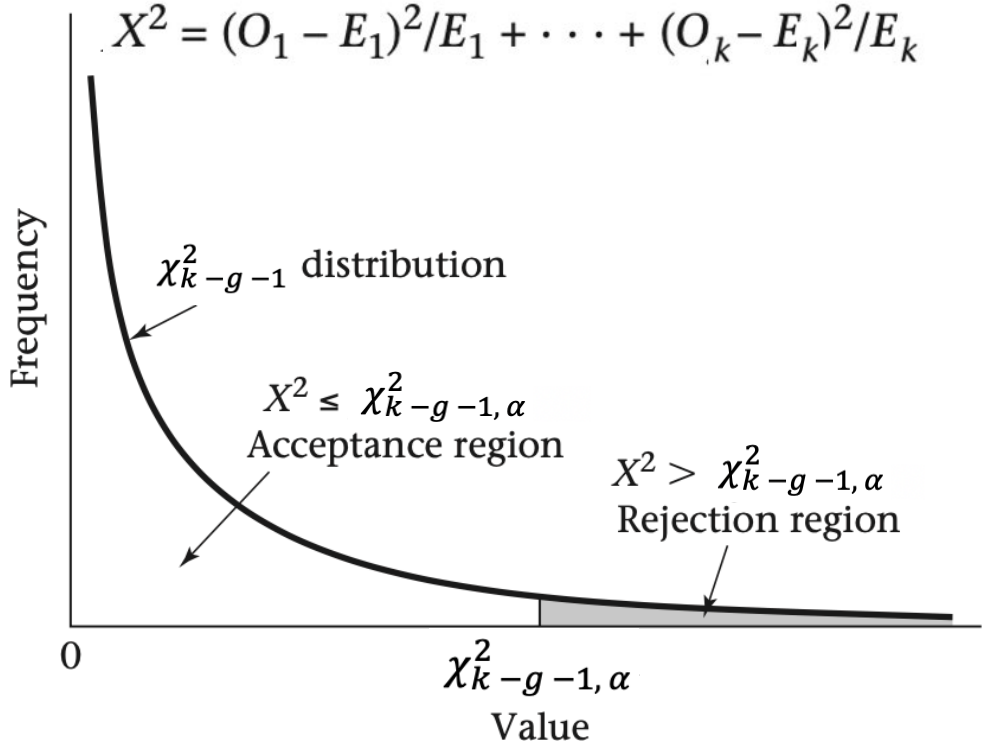
$$\sum_{i=1}^{k} \frac{(X_i - np_i(\hat{\theta}))^2}{np_i(\hat{\theta})} \xrightarrow{D} \chi^2 (k - 1 - \dim \Omega)$$

$\dim \Omega$: the dimension of the set Ω. For our purpose, let's roughly understand it as the number of parameters.

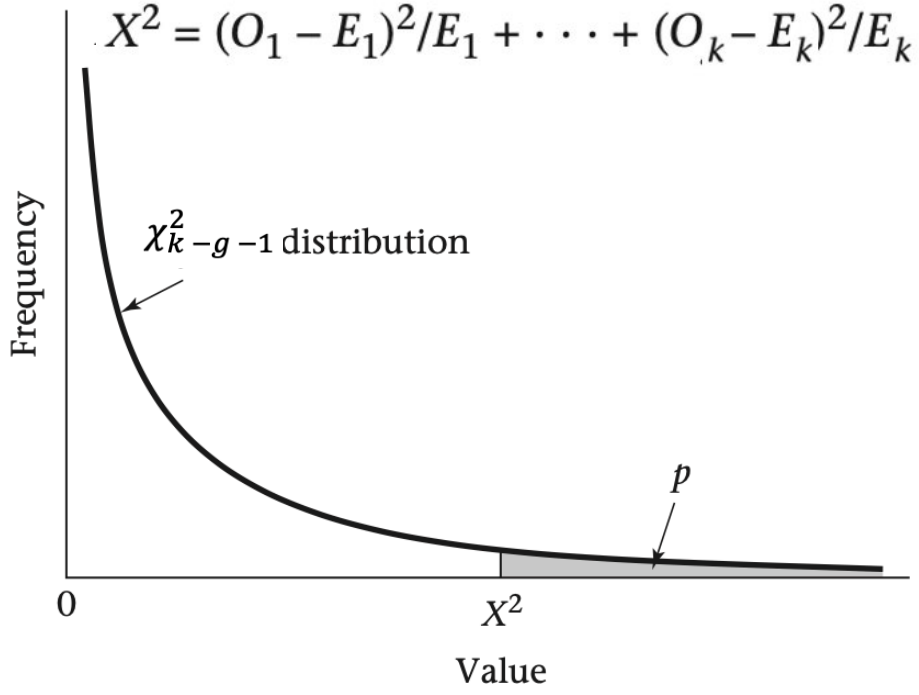
Goodness of fit when cell probabilities depend on an **unknown** parameter

k: number of categories
g: parameter degree of freedom

Acceptance and rejection regions for the chi-square goodness-of-fit test



Computation of the *p*-value for the chi-square goodness-of-fit test



Goodness of fit when cell probabilities depend on an **unknown** parameter

Table 10.22 **Frequency distribution of mean diastolic blood pressure for adults 30–69 years old in a community-wide screening program in East Boston, Massachusetts**

Group (mm Hg)	Observed frequency	Expected frequency	Group	Observed frequency	Expected frequency
<50	57		≥80, <90	4604	
≥50, <60	330		≥90, <100	2119	
≥60, <70	2132		≥100, <110	659	
≥70, <80	4584		≥110	251	
			Total	14,736	

$\bar{x} = 80.68,\ s = 12.00$

H_0: *The distribution is normal*

The expected frequency within a group interval from a to b would then be given by

$$14{,}736\{\Phi[(b-\mu)/\sigma]-\Phi[(a-\mu)/\sigma]\}$$

Thus the expected frequency within the (≥50, < 60) group would be

$$14{,}736\times\{\Phi[(60-80.68)/12]-\Phi[(50-80.68)/12]\}$$
$$=14{,}736\times[\Phi(-1.723)-\Phi(-2.557)]$$
$$=14{,}736\times(.0424-.0053)=14{,}736(.0371)=547.1$$

$$X^2=(O_1-E_1)^2/E_1+\cdots+(O_8-E_8)^2/E_8$$
$$=(57-77.9)^2/77.9+\cdots+(251-107.2)^2/107.2=350.2\sim\chi^2_5 \text{ under } H_0$$

$\chi^2_{5,0.001} = 20.52 < 350.2$

Null is rejected.

Goodness of fit when cell probabilities depend on an **unknown** parameter

H_0: X is Poisson

MLE: $\hat{\lambda} = \bar{x} = 5.4$

7	4	3	6	4	4	5	3	5	3
5	5	3	2	5	4	3	3	7	6
6	4	3	11	9	6	7	4	5	4
7	3	2	8	6	7	4	1	9	8
4	8	9	3	9	7	7	9	3	10

A_1 = {0,1,2,3}, A_2 = {4}, A_3 = {5}, A_4 = {6}, A_5 = {7}, and A_6 = {8,9,10,...}

	A_1	A_2	A_3	A_4	A_5	A_6
Frequency	13	9	6	5	7	10
Probability	0.213	0.160	0.173	0.156	0.120	0.178
Expected ($50p_i$)	10.65	8.00	8.65	7.80	6.00	8.90

$$q_5 = \frac{[13 - 50(0.213)]^2}{50(0.213)} + \cdots + \frac{[10 - 50(0.178)]^2}{50(0.178)}$$

$$= 2.763 < 9.488 = \chi^2_{0.05}(4),$$

H_0 is not rejected at the 5% significance level.